

Sessional Exam - II.
(Solution)

Q1: Consider the events: H: Husband watches a certain show
W: Wife watches the show.

(a)

$$\begin{aligned} P(W \cap H) &= P(W) \cdot P(H|W) \\ &= (0.5)(0.7) \\ &= 0.35 \end{aligned}$$

$$b) P(W|H) = \frac{P(W \cap H)}{P(H)} = \frac{0.35}{0.4} = 0.875$$

$$\begin{aligned} c) P(W \cup H) &= P(W) + P(H) - P(W \cap H) \\ &= 0.5 + 0.4 - 0.35 \\ &= 0.55 \end{aligned}$$

Q2: Let "X" denote number of heads.

a)

X	P(X=x)
0	$\frac{1}{8}$
1	$\frac{3}{8}$
2	$\frac{3}{8}$
3	$\frac{1}{8}$
	1

$$b) E(X) = \sum x P(X=x)$$

$$\begin{aligned} &= 1 \times \frac{3}{8} + 2 \times \frac{3}{8} + 3 \times \frac{1}{8} \\ &= \frac{3}{8} + \frac{6}{8} + \frac{3}{8} \\ &= \frac{3}{2} \end{aligned}$$

$$c) V(X) = E(X^2) - [E(X)]^2$$

$$E(X^2) = \frac{3}{8} + \frac{12}{8} + \frac{9}{8} = 3$$

$$V(X) = \frac{3}{1} - \left(\frac{3}{2}\right)^2 = \frac{3}{4}$$

Q3: a) $\int_0^1 f(x) dx = 1 \Rightarrow k \int_0^1 x^{\frac{1}{2}} dx = 1$

$$k \frac{x^{\frac{3}{2}}}{\frac{3}{2}} \Big|_0^1 = 1$$

$$k = \frac{3}{2}$$

b) $P(X \leq u) = \frac{3}{2} \int_0^u x^{\frac{1}{2}} dx = \frac{3}{2} \cdot \frac{x^{\frac{3}{2}}}{\frac{3}{2}} \Big|_0^u$

$$= x^{\frac{3}{2}}$$

c) $P(0.3 < X < 0.6) = \Phi(0.6) - \Phi(0.3)$

$$= (0.6)^{\frac{3}{2}} - (0.3)^{\frac{3}{2}}$$

$$= 0.3004$$

$$\underline{\underline{Q4}} a) P(X \geq 3) = 1 - P(X \leq 2)$$

$$= 1 - [P(X=0) + P(X=1) + P(X=2)]$$

$$= 1 - \left[\binom{5}{0} (0.30)^0 (0.70)^5 + \binom{5}{1} (0.30)^1 (0.70)^4 + \binom{5}{2} (0.30)^2 (0.70)^3 \right]$$

$$= 0.16308$$

$$b) P(X \geq 2) = 1 - P(X \leq 1)$$

$$= 1 - \{P(X=0) + P(X=1)\}$$

$$= 1 - \left[e^{-3} \frac{3^0}{0!} + e^{-3} \frac{3^1}{1!} \right]$$

$$= 0.8009$$

$$\underline{\underline{Q5}}: a) P(X < 91) = P\left(\frac{X-4}{\sigma} < \frac{91-73}{8}\right) = P(Z < 2.25)$$

$$= \underline{\underline{0.9878}}$$

or

$$b) P(X > 60) = P\left(\frac{X-4}{\sigma} > \frac{60-73}{8}\right) = P(Z > -1.625)$$

$$= 1 - P(Z < -1.625)$$

$$= \underline{\underline{0.9479}}$$

$$c) P(60 < X < 91) = P(-1.625 < Z < 2.25)$$

$$= \Phi(2.25) - \Phi(-1.625)$$

$$= \underline{\underline{0.9357}}$$